

Menses Mortis

Fight against a never-ending cycle.

Game Design Document

Concept, Systems & Adaptive AI Architecture

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Engine: Unity (C#) | Genre: 2D Grid Roguelike | Solo Project

1. Overview

Menses Mortis is a two-dimensional, grid-based roguelike built in Unity in which the central pillar is not difficulty scaling but adaptation: the enemies study the player and learn to counter that individual's tendencies over time. The guiding question behind the project is a simple inversion of how players normally relate to games. Players spend their time studying enemy patterns; this game asks what it would feel like if the enemies studied the player back.

The game is set inside the female reproductive system across the twenty-eight-day menstrual cycle. The player controls a humanized blood clot that has grown legs and is attempting to dig its way out of the uterus and through the vaginal canal to the exit. The adaptation mechanic is not a technical trick bolted onto an unrelated setting; it is biologically motivated. The body has genuine adaptive systems, most notably the immune response, which observes what it encounters, remembers it, and answers faster and harder the next time. The enemy's learning the player is a direct expression of that biology. In this project, the theme and the mechanic are the same idea, and every design decision is measured against that unity.

The headline capability is stated plainly: *an adaptive NPC system that learns player behavior over time and dynamically adjusts its strategy to counter individual playstyles.*

2. Creative Vision & Tone

The tone of Menses Mortis is ironic horror with comedic relief. It is a horror premise treated with affection rather than solemnity. The dread is real, the endurance is real, and the humor is real, and the design requires all three registers to coexist. The game is never grim to the exclusion of levity, and never jokey to the exclusion of weight. This balance reflects a lived attitude toward its subject: the experience it draws from is frustrating and relentless, but people who live it also laugh about it, because refusing to be ground down by this monthly cycle is a form of strength.

A strict point-of-view rule governs the entire project. The player is the blood clot, resulting in never being given a gender. The interior of the body belongs to a woman, but that is the limited entirety of her presence in the game. What she endures is never described to the player. It is instead conveyed through the systems the player is subjected to:

- A clock that only moves forward,
- A hormonal tide that can be watched but never controlled
- A body that learns and counters
- Slow petrification under damage
- An exit that ultimately only returns the player to the beginning. The player undergoes the experience rather than being told about it.

From this follows the project's central writing constraint: show, never tell. There is no expository text explaining menstruation, no narration about womanhood, and no dialogue from the woman. If a feeling cannot be delivered through mechanics, environment, or a very small number of tone-critical lines, it does not belong in the game. Only two moments speak to the player directly, the death screen and the closing message, and they earn that directness precisely because by the time the player reads them, they have already lived the loop the words refer to.

The difficulty is unchosen and uneven by design. That frustration is the payload, not a balance problem to be solved.

This intent explains why several systems in this document resist the usual instinct to smooth a game out. The hormonal difficulty curve is deliberately uneven. The clock refuses to rewind on death. The player cannot mitigate the pressure they can see building. These are not oversights; they are the experience the game exists to deliver.

3. Design Pillars

Five pillars act as decision filters throughout development. Any proposed mechanic that fails one of them is reconsidered or cut.

- **Concept first.** Every mechanic must earn its place by serving the theme. Nothing is added for purely mechanical reasons.
- **Scientific honesty over mechanical convenience.** Where the biologically accurate version and the convenient version disagree, accuracy wins, because accuracy is the point of the project.
- **A precise emotional target.** The intended feeling is something that makes the player uncomfortable, fused with the sense of one's own body turning against oneself.
- **Reveals stay vague.** The player should feel watched and studied, but is never handed a readable strategy printout of what the enemies know.
- **Single parameterized systems over branching builds.** Difficulty modes are dials on one system rather than separate code paths, which keeps the game honest and maintainable.

4. Visual Style & Art Direction

The art direction is still evolving and this section captures the current creative direction rather than a locked specification. The overall feeling the game reaches for is uncanny. The world is the interior of a living body rendered as a place with its own sentience, where the terrain, the enemies, and the environment all read as cells and tissue that are aware, in some low and unsettling way, of the intruder moving through them. The player should never quite lose the

sense that the level itself is watching, which pairs directly with the adaptive AI: the body does not merely contain enemies, the body itself is the enemy.

Mechanically, the game sits in the lineage of the classic digger, in the family of *Dig Dug* by Namco in 1982, and *Mr. Driller*, also by Namco in 1999, where the player carves downward through destructible terrain. The current visual reference point is the presentation of *Ball x Pit* by Kennie Sun, whose look and feel are the closest existing touchstone for the density, energy, and readability the project is aiming at. It is worth stating clearly that this is a reference for presentation only; *Menses Mortis* is its own game, and the digger heritage is something the art direction will consciously decide to embrace or push against rather than inherit by default.

The intended play sensation is quick, responsive, sprite-based action with a *Metroidvania* feel. Movement and combat should feel snappy and immediate, with fast sprite action rather than deliberate, weighty motion, so that the moments where the player *does* slow down carry meaning. This is important because sluggishness is a designed state in this game rather than an accident of feel: as the player takes damage, they calcify and move more heavily, so a baseline of quick, crisp responsiveness is what makes that gradual loss legible and frightening.

One constraint shapes every visual choice: the world rotates. Because the gravity-flip mechanic can reorient the entire play-field by ninety degrees in either direction, and occasionally by a full half-turn, the visual language must remain clear and readable in any orientation. Any style the project adopts has to survive being turned on its side without becoming confusing, which rules out art direction that depends on a single fixed sense of up and down.

5. World & Core Loop

The player character is a humanized blood clot with legs, digging through hardened old blood that forms the game's destructible terrain. That terrain is arranged on a grid and is tiered by durability. The permanent boundary of the world is the wall of the vaginal canal, which can never be broken under any condition and defines the edges of play. Enemies enter the world by spawning from the cervix, and the objective is to reach the exit at the far end of the canal.

The core loop is straightforward to state and rich in consequence. The player digs toward the exit; enemies move to interfere; the player kills them with a ranged pellet weapon, which is also the act that teaches the body; the player takes damage and gradually petrifies; the player eventually dies and respawns, with the cycle clock unchanged; and the loop repeats until the canal is finally cleared. The crucial detail is that killing enemies is the mechanism by which the enemies learn, so the player's primary tool for survival is also the primary way the player makes the world harder for themselves. That tension sits at the heart of the game.

6. The Gravity-Flip System

The world is subject to a gravity-flip mechanic that reorients everything within it. Flips rotate the play-field ninety degrees to the left or right, and on rarer occasions a full one hundred and eighty degrees. When a flip occurs, it affects the player, the enemies, the terrain blocks, and any loose items, all of which are subject to the new direction of gravity. A small set of elements are deliberately exempt: anchor blocks that serve as fixed orientation landmarks, the indestructible canal walls, and the shops.

A rule of particular importance is that when the world flips, every enemy's position *and* its movement orientation update to the new configuration. No enemy is exempt from the flip. This is treated as non-negotiable and is verified explicitly in testing, because an enemy that failed to reorient would break both the fairness and the internal logic of the world. The frequency of flips is not fixed; it is driven by the hormonal state of the body described in the next sections, so the world becomes more volatile precisely when the body is most agitated.

In the current design, flips occur automatically according to that hormonal pressure. Allowing the player to trigger flips deliberately is noted as a possible future mode but is not part of the core experience, which intentionally denies the player control over the world's orientation.

7. The Two Clocks

Two independent timekeeping systems run in parallel and are kept strictly decoupled. Conflating them would collapse two distinct sources of pressure into one, so the design treats them as separate throughout.

7.1 The Cycle Clock

The cycle clock advances through the four phases of the menstrual cycle in their correct medical order: the menstrual phase, the follicular phase, ovulation, and the luteal phase, after which it loops back to the menstrual phase. The order and the names are fixed and are drawn from the real biology of the cycle. The menstrual phase, roughly days one through five, is when the lining sheds and both estrogen and progesterone sit at their lowest. The follicular phase, spanning roughly the first thirteen days, sees estrogen rise while the lining thickens and follicles mature. Ovulation, around day fourteen, is marked by the surge in luteinizing hormone that releases a mature egg and represents the sharpest hormonal peak of the cycle. The luteal phase, roughly days fifteen through twenty-eight, sees progesterone rise and then crash together with estrogen, which triggers the return of menstruation.

The defining rule of the cycle clock is that death does not reset it. If the player dies during ovulation, they respawn still in ovulation, and the clock continues to advance from where it was. The body does not rewind. This reframes death itself: dying does not cost the player the cycle, it

costs them ground inside a clock that never waited for them. The clock advances only during active play and stops when the game is closed, so the cycle is a pressure that ticks only while the player is inside it. Finally, the document notes an honest simplification: in reality the follicular phase overlaps the menstrual phase, since both begin on day one, but the interface presents four clean sequential segments for legibility. This is a deliberate design choice in service of clarity, not an error in the underlying biology.

7.2 The Learning Clock

The learning clock is the body's accumulating memory of the player, and it is entirely separate from the cycle clock. Where the cycle clock is a fixed biological calendar, the learning clock grows in response to how the player actually behaves. Its full architecture is the subject of Section 9.

8. The Hormone System

Hormone level in *Menses Mortis* is modeled as a continuous value that changes day by day, rather than a set of four fixed levels attached to the four phases. This choice is central rather than cosmetic. Hormone levels in the real cycle are not flat, and the unevenness of that curve is exactly the sensation the game intends to convey. The value is low and flat during the menstrual phase, climbs steadily through the follicular phase, spikes sharply and briefly at ovulation with the surge in luteinizing hormone, and then runs high before crashing back toward baseline across the luteal phase. Late in the follicular phase the game is measurably tenser than early in it, and the ovulation spike is a genuine spike rather than a gentle rise.

The phase and the hormone value are treated as two separate inputs to the game. The phase determines which enemy types are permitted to appear, since each phase has its own roster. The hormone value, independently, determines how intense the moment is: how many enemies are present, how quickly they move, whether they take their larger and more aggressive form, and how often the world flips. Two moments in different phases might share a similar level of hormonal heat while drawing on completely different rosters of enemies, which keeps the two systems from becoming redundant. Speed scaling is applied per enemy type rather than uniformly, so a rising hormone level does not accelerate every enemy by the same amount.

The player is made continuously aware of this value through the interface, but is given no power over it. This read-only relationship to the hormonal tide is a deliberate expression of the game's central theme, and is discussed further in the section on the user interface.

9. Adaptive AI: The Centerpiece

The adaptive artificial intelligence is the reason the project exists, and its design is driven end to end by the game's established rules rather than by a desire to use any particular technique. The requirements are concrete. The system must build a per-type profile of the player's tendencies; it must remain legible enough to explain clearly in an interview; it must support a residue of learning that survives a partial reset; it must distinguish long-term habits from short-term reactions; it must be tunable through a single decay parameter; and it must be buildable and debuggable by a solo developer rather than a research team. These requirements, taken together, effectively select the architecture.

9.1 Why Not Reinforcement Learning

Reinforcement learning was considered and set aside, and being able to explain that decision is itself part of the portfolio value. Reinforcement learning is powerful but opaque, slow to converge without large volumes of play data, and difficult to tune predictably. In a fast roguelike where individual enemies are killed within seconds, the volume of clean training data required never materializes, and an opaque model works directly against a game whose emotional effect depends on legibility and careful balance. A transparent, weighted system, by contrast, can be inspected, explained, and tuned, and it produces the readable per-player profile the design calls for. Reinforcement learning is therefore documented as a possible future extension rather than the core approach, and the architecture is deliberately built so that it could be added later without disruption.

9.2 A Three-Layer Architecture

The system is separated into three layers with clear responsibilities. The observation layer runs constantly, recording the player's movement direction, positional preferences, reactions after a world flip, and short sequences of actions, organized per enemy type and per phase. It only ever watches and never acts. The decision layer, which can be thought of as a swappable brain, exposes a single interface that returns a directional movement bias together with an optional instruction to halt. Multiple brains implement that interface: a naive brain that returns no bias and serves as a baseline and testing tool, a weighted-profile brain that ships first, a Markov brain layered on afterward, and a reserved slot for a future reinforcement-learning brain that is documented but not built. The execution layer is a behavior tree, one per enemy type, that takes whatever bias the current brain produces and carries it out within that enemy's legal movement vector. Crucially, the behavior tree never knows which brain is driving it, which is what makes the brain genuinely swappable and, incidentally, what makes the system easy to test.

9.3 The Learning Model

Each enemy type maintains two parallel sets of weights across the eight directions of movement. A slow set of weights, with a small learning rate and high inertia, captures the player's deep and consistent habits; these slow weights constitute the residue that survives a reset. A fast set of

weights, with a large learning rate and low inertia, captures recent and situational tendencies. The bias the enemy ultimately acts on blends the two, with the fast weights given more influence for responsiveness. This pairing is the mechanism that lets the game distinguish a player's ingrained habits from their momentary reactions, and it turns out to be the same mechanism that makes memory decay simple to implement.

9.4 Observe, Then Act on Death

A single rule governs when learning becomes visible to the player. The body observes continuously and builds its profile of the player at all times, but an enemy type is only permitted to act on that profile after one of its own has died to the player's pellets. Until that first death, the type remains naive no matter how much the body has quietly observed. Death is the trigger that flips a type from watching to acting on what the body already knew. This rule is faithful to the immune-system metaphor, in which the body learns from what it loses to a pathogen, and it gives the player a genuine, if unsettling, agency over their own difficulty: every kill is a lesson the player chose to teach.

9.5 What Adaptation Looks Like in Play

Adaptation is expressed only through movement, and only after an enemy has spawned. An activated enemy biases its movement toward the regions the player tends to occupy, so a player who favors the right side of the field will find activated enemies drifting to meet them there. Activated enemies can also halt deliberately to block the player's path rather than continuing to drift along their fixed vector. Once the Markov layer is active, enemies additionally pre-position for the player's likely next action rather than merely for their favorite location. A strict fairness rule accompanies all of this: learned knowledge never influences where enemies spawn. Spawn placement is governed only by the phase roster and the hormone value, so the player is always presented with a fair starting board, and adaptation shapes the encounter only as it unfolds.

9.6 Memory, Decay, and Difficulty

Within a single phase, the body's memory persists across the player's deaths, so dying does not wipe what the enemies have learned. When a full cycle completes and a phase comes around again, the fast weights are cleared while the slow weights are retained. The effect is a returning phase that is harder than it was on first encounter but easier than it was at its peak; pressure builds within a phase and is partially released at the turnover, producing a breathing rhythm that mirrors the body's own. Because this decay is simply the clearing of the fast weights and the retention of the slow ones, difficulty reduces to a single parameter. In the standard mode the decay is active. In a super-hard mode the decay rate is set to zero and both sets of weights are retained, so the body never forgets. This is one parameterized system rather than two separate code paths.

9.7 Fairness Caps

A weighted, learning system will, left unchecked, converge toward near-perfect prediction of a consistent player, which would make long sessions feel unfair. Four caps are therefore treated as foundational rules rather than as later balance adjustments. A cap on the maximum bias magnitude ensures that even a fully-trained enemy still moves incorrectly a meaningful fraction of the time, which guarantees the existence of permanently exploitable gaps. A minimum-observations threshold before a type may act prevents wild overcorrection from thin data. Deliberate prediction noise keeps the Markov layer from ever becoming deterministic, so the enemy reads as an opponent rather than as a system that sees through walls. And a saturation ceiling on the slow weights protects the decay mechanic across very long playthroughs, since without it the residue floor would eventually rise to near-maximum and decay would stop mattering. All four caps remain in force even in the super-hard mode, because that mode disables decay only; a body that never forgets is not the same as a body that is omniscient. Adaptation is thus designed to climb steeply and then flatten at a chosen ceiling, never to climb without limit. Every cap and learning rate is exposed as a tunable field rather than hardcoded, because the correct values can only be found through playtesting.

9.8 The “Will Remember That” Beat

When the player kills an enemy, and only when that kill actually taught the body something new, a brief message rises from the location of the death, naming the enemy and stating that it will remember what just happened. The moment borrows the cadence of the well-known narrative prompt in which a character will remember an action, and turns it against the player. It fires on a kill rather than on a death, it is rationed to genuine new learning so that it never becomes wallpaper, and it never reveals specifics. Its power lies in the inversion it dramatizes: the player won the exchange, and the game replies that winning is precisely what fed the memory.

10. Enemies

Enemy movement is built on a chess-piece model that keeps the underlying behavior legible and grid-honest. A naive enemy moves along a fixed vector: some move horizontally like a rook confined to one axis, some move vertically in the same way, and some move diagonally like a bishop, in either a left-leaning or a right-leaning variant. This constrained vocabulary of motion is what makes the later adaptation readable, because a bias applied to a simple, predictable vector is something the player can actually perceive and reason about.

Each phase draws on its own roster of enemies, and those rosters never cross between phases, which both reinforces the player's sense of where they are in the cycle and neatly compartmentalizes the body's learning. The menstrual phase presents only horizontal and vertical movers and is the simplest, lowest-pressure ground on which the player begins. The follicular

phase introduces diagonal movement as a new threat vector while the body builds. Ovulation brings all three movement types together and is the hottest and most aggressive phase, home to the largest number of enemies in their more dangerous form. The luteal phase also fields all types, but its pressure comes less from crowding the screen and more from a heightened frequency of world flips, so the environment itself becomes the primary source of threat.

Aggression is driven by the hormone value rather than the phase name. A baseline enemy occupies a single grid square and moves at standard speed. An aggressive enemy, emerging at higher hormone levels, occupies two grid squares and moves faster, with the exact increase depending on the enemy type. Enemy names in the current build are placeholders, tied loosely to the hormones they evoke and kept comprehensible rather than clinically precise, and are stored as editable data so that they can be renamed freely without touching code. Humor is welcome in this naming.

11. Combat & Damage

The player's primary weapon is a pellet, and it serves two roles at once: it breaks terrain and it kills enemies. Pellets are finite but recharge on their own after a short delay of one to two seconds, so no pickup is required to keep firing, and the detailed scaling of the weapon is deliberately deferred until after the prototype is playable. The dual purpose of the pellet is meaningful, because the same action that clears a path also produces the kills that teach the body, binding the player's progress and the player's growing difficulty into a single verb.

The terrain the player digs through grows tougher the deeper they go, which is to say the closer they get to the exit. This durability scales with depth alone and is unaffected by the phase, and the player never needs a tool upgrade to break any block; harder blocks simply take more hits. The effect is a steadily mounting resistance that approaches the exit feel like a genuine push against an increasingly reluctant body.

Damage is modeled as petrification. Every hit an enemy lands drains the player's life, and as life falls, the player moves more sluggishly, calcifying gradually toward stone. Death is not an explosion; instead, the player hardens into one of the blood clots that make up the terrain. The consequence is quietly horrifying and thematically exact. The level is literally built out of fallen versions of the player; every death makes the world a little more solid, and death itself is simply the final frame of a petrification that every single hit has been advancing all along. The enemies' entire project, the thing their adaptation serves, is to turn the player into the terrain, and to win is to refuse to become a wall.

12. Progression & Economy

Progression follows a Metroidvania-inflected structure in which the player begins every run already equipped with a starter weapon. That starter pellet occupies a protected safe slot and is always retained through death, so the player is never left defenseless at the start of a run. Other carried items occupy loss slots, whose contents are dropped into the world on death and can be recovered by returning to them, which gives death a tangible cost without making it total.

An economy runs alongside this. Rare blocks encountered while digging can be sold for currency, and that currency is spent at shops that offer upgrades, status effects, and expansions to the player's inventory. The shops occupy a deliberately special place in the world: they spawn within the rotating walls and are exempt from the physics that reorient everything else, standing as fixed, stable islands in a world that keeps flipping around them. This exemption is an incentive as much as a convenience, drawing the player to rush toward a shop or to pass through one while the surrounding world remains in flux.

13. Death & Win Conditions

The death screen is one of the only two moments in the game that speak plainly to the player, and it is written to unsettle rather than to console. On death the edges of the screen darken and close inward, the frame narrowing until only the player, now hardened into terrain, remains highlighted. A line of encouragement appears, addressing the player by name and telling them they can do this, and then the screen snaps briskly back to the starting area. The horror is ironic: the words are supportive while the framing is anything but, and the abruptness of the return, with no ceremony and no pause, states without saying so that the player is already going again.

The path to victory runs the fixed length of the canal, gated by terrain that grows more durable with depth. At the exit there is no final block but a final enemy, and pointedly it is not a part of the body; it is a foreign hormone, the very thing that gave the player life, and the game's small deliberate story beat is delivered in this encounter. Defeating it drops loot and upgrades and opens the exit, and the player passes out of the canal, into a toilet. A closing message then lands the game's real subject. Its phrasing is still being refined, but its meaning is settled: that the player will have to do this again, that the cycle is not an event but an ordinary recurring fact of a woman's life, returning roughly every twenty-eight days, month after month, until menopause. The line is written to land as respect and endurance, never as a joke, and it earns its directness because by the time the player reads it they have lived the loop it describes.

14. User Interface

The interface keeps the player continuously oriented within the cycle. Across the top of the screen, always visible, sit the current day, the current phase, and the phase to come. In a corner near this phase clock stands a small vertical bar that reads out the hormone level, filling from

empty at the bottom to full at the top in the manner of a battery. Because the underlying hormone value is continuous, this bar is never static; it breathes slowly upward and downward from day to day, so the player can watch pressure gather well before it arrives and learn to dread the climb toward the ovulation spike.

The hormone bar is strictly read-only. The player cannot act on it, spend against it, or blunt it in any way. It is a meter of dread rather than a control panel, and this powerlessness is the intended experience: watching the storm approach and being unable to stop it is exactly the feeling the game is built to produce. Separately, a developer-facing debug overlay visualizes the learned weights for each enemy type in real time. This overlay is not optional; it is at once the essential instrument for tuning the adaptive system and the clearest way to demonstrate that system to others, which makes it a core part of the project's value as a portfolio piece.

15. Technical Approach & Build Order

The game is built in Unity in C#, and development follows a strict order in which each step must be demonstrably working before the next begins. The world's foundations come first: the grid, player movement, and the gravity-flip system, followed by terrain blocks with depth-scaled durability and the pellet firing and recharge mechanic. Enemies arrive next in their naive, fixed-vector form, at which point the correct reorientation of every enemy through a world flip is verified explicitly. Damage, the petrification of movement, death into terrain, and respawn are then implemented, followed by the cycle clock, the continuous hormone value, and the associated interface elements.

Only once that foundation is solid does the adaptive system go in, and it too is layered deliberately. The brain interface, the naive brain, and the behavior-tree execution layer come first, then the observation layer, which at this stage records without having any effect on behavior. The weighted-profile brain follows, complete with its paired fast and slow weights, all four fairness caps, and the activate-on-death rule. Decay at cycle turnover and the difficulty parameter come next, then the Markov layer composed on top, and then the debug overlay that visualizes the learned weights. The unsettling kill message is added after the learning it depends on exists. The economy, with its shops, rare-block currency, and inventory slots, and finally the boss encounter, the win condition, and the death and closing screens complete the build. This ordering ensures that a testable, playable skeleton exists early and that the adaptation, the project's most ambitious element, is layered onto a stable base rather than entangled with it.

A note on architecture underlies the whole plan: every tunable value is exposed as an editable field rather than hardcoded, and the brain interface is kept genuinely swappable. Together these choices make the game tunable through playtesting, straightforward to demonstrate, and open to

a more sophisticated learning approach in the future without requiring the rest of the game to change.

16. Portfolio Positioning

As a portfolio piece, Menses Mortis is designed to demonstrate systems thinking can be an integral part of storytelling, rather than the mere application of a fashionable technique. Its headline is an adaptive enemy system that learns individual player behavior over time and adjusts its strategy to counter it, and the strength of the project lies in the coherence between that system and everything around it. The adaptation is not decorative; it is motivated by the body's real biology, expressed through legible movement, bounded by explicit fairness rules, and tunable through a single honest parameter. The deliberate decision to favor a transparent weighted-and-Markov approach over reinforcement learning, and the ability to articulate why that decision was correct for these particular constraints, is itself the kind of judgment the project is meant to showcase. The result is interactive, demonstrable through its own debug tooling, and grounded in a concept that gives every mechanic a reason to exist.